GenQuantis Discovery Platform

Advanced Molecular Analysis & Formulation Report

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THERAPEUTIC TARGET DISCOVERY REPORT

Target Name:	Epidermal growth factor receptor
UniProt ID:	P00533
PDB Structures:	9GC4, 7ZYM, 9DF3, 5LV6, 6P8Q, 5J9Z, 9HBO, 5FED, 5UG9, 3BUO, 5EM5, 4J
Source:	RCSB PDB (Crystal)

Genomic Sequence Analysis

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MRPSGTAGAALLALLAALCPASRALEEKKVCQGTSNKLTQLGTFEDHFLSLQRMFNNCEVVLGNLEITYVQRNYDLSFLKTIQEVAGYVLIALNTVER
IPLENLQIIRGNMYYENSYALAVLSNYDANKTGLKELPMRNLQEILHGAVRFSNNPALCNVESIQWRDIVSSDFLSNMSMDFQNHLGSCQKCDPSCPN
GSCWGAGEENCQKLTKIICAQQCSGRCRGKSPSDCCHNQCAAGCTGPRESDCLVCRKFRDEATCKDTCPPLMLYNPTTYQMDVNPEGKYSFGATCVKK
CPRNYVVTDHGSCVRACGADSYEMEEDGVRKCKKCEGPCRKVCNGIGIGEFKDSLSINATNIKHFKNCTSISGDLHILPVAFRGDSFTHTPPLDPQEL
DILKTVKEITGFLLIQAWPENRTDLHAFENLEIIRGRTKQHGQFSLAVVSLNITSLGLRSLKEISDGDVIISGNKNLCYANTINWKKLFGTSGQKTKI
ISNRGENSCKATGQVCHALCSPEGCWGPEPRDCVSCRNVSRGRECVDKCNLLEGEPREFVENSECIQCHPECLPQAMNITCTGRGPDNCIQCAHYIDG
PHCVKTCPAGVMGENNTLVWKYADAGHVCHLCHPNCTYGCTGPGLEGCPTNGPKIPSIATGMVGALLLLLVVALGIGLFMRRRHIVRKRTLRRLLQER
ELVEPLTPSGEAPNQALLRILKETEFKKIKVLGSGAFGTVYKGLWIPEGEKVKIPVAIKELREATSPKANKEILDEAYVMASVDNPHVCRLLGICLTS
TVQLITQLMPFGCLLDYVREHKDNIGSQYLLNWCVQIAKGMNYLEDRRLVHRDLAARNVLVKTPQHVKITDFGLAKLLGAEEKEYHAEGGKVPIKWMA
LESILHRIYTHQSDVWSYGVTVWELMTFGSKPYDGIPASEISSILEKGERLPQPPICTIDVYMIMVKCWMIDADSRPKFRELIIEFSKMARDPQRYLV
IQGDERMHLPSPTDSNFYRALMDEEDMDDVVDADEYLIPQQGFFSSPSTSRTPLLSSLSATSNNSTVACIDRNGLQSCPIKEDSFLQRYSSDPTGALT
EDSIDDTFLPVPEYINQSVPKRPAGSVQNPVYHNQPLNPAPSRDPHYQDPHSTAVGNPEYLNTVQPTCVNSTFDSPAHWAQKGSHQISLDNPDYQQDF
FPKEAKPNGIFKGSTAENAEYLRVAPQSSEFIGA
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PPI Interactome Mapping (STRING)

Partner	Method	Interaction Type	String Score
TP53	AP-MS	Physical Interaction	98%
MDM2	Y2H	Regulatory Binding	95%
ATM	Biochemical Assay	Phosphorylation	88%
KAT5	AlphaFold-Multimer	Structural Complex	72%

Identified ChEMBL Binders (Top 15)

ChEMBL ID	Metric	Value	SMILES Signature
CHEMBL68920	IC50	41.0 nM	Cc1cc(C)c(/C=C2\C(=O)Nc3ncnc(Nc4ccc(F)c(Cl)c4)c32)[nH]1...

CHEMBL68920	IC50	300.0 nM	Cc1cc(C)c(/C=C2\C(=O)Nc3ncnc(Nc4ccc(F)c(Cl)c4)c32)[nH]1...
CHEMBL68920	IC50	7820.0 nM	Cc1cc(C)c(/C=C2\C(=O)Nc3ncnc(Nc4ccc(F)c(Cl)c4)c32)[nH]1...
CHEMBL69960	IC50	170.0 nM	Cc1cc(C(=O)N2CCOCC2)[nH]c1/C=C1\C(=O)Nc2ncnc(Nc3ccc(F)c...
CHEMBL69960	IC50	40.0 nM	Cc1cc(C(=O)N2CCOCC2)[nH]c1/C=C1\C(=O)Nc2ncnc(Nc3ccc(F)c...
CHEMBL69960	IC50	440.0 nM	Cc1cc(C(=O)N2CCOCC2)[nH]c1/C=C1\C(=O)Nc2ncnc(Nc3ccc(F)c...
CHEMBL137635	IC50	9300.0 nM	CN(c1ccccc1)c1ncnc2ccc(N/N=N/Cc3ccccn3)cc12...
CHEMBL443268	IC50	5310.0 nM	Cc1cc(C(=O)NCCN2CCOCC2)[nH]c1/C=C1\C(=O)N(C)c2ncnc(Nc3c...
CHEMBL76589	IC50	125.0 nM	N#CC(C#N)=C(N)/C(C#N)=C/c1ccc(O)cc1...
CHEMBL304271	IC50	0.45 nM	CCN(CC)CC(O)CNC(=O)c1cc(C)c(/C=C2\C(=O)Nc3ncnc(Nc4ccc(F...
CHEMBL296407	IC50	10000.0 nM	N#C/C(=C\c1ccc(O)c(O)c1)C(N)=O...
CHEMBL296407	Ki	3500.0 nM	N#C/C(=C\c1ccc(O)c(O)c1)C(N)=O...
CHEMBL310798	IC50	3000.0 nM	N#CC(C#N)=Cc1cc(O)c(O)c(O)c1...
CHEMBL310798	Ki	1000.0 nM	N#CC(C#N)=Cc1cc(O)c(O)c(O)c1...
CHEMBL135592	IC50	8000.0 nM	C/N=N/Nc1ccc2ncnc(N(C)c3ccccc3)c2c1...

Platform Disclaimer: This report is generated by GenQuantis Discovery AI. Structural and bioactivity data is retrieved from external repositories (RCSB, UniProt, ChEMBL, STRING). Experimental validation is required for target tractability.